Year 13 Maths - Numerical solution of equations

Similar questions - Iteration and convergence

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The equation $x^3 + 3x - 7 = 0$ has a single real root α.

(a) Show that α lies between 1 and 2. **[2]**

(b) Show that the equation can be rearranged as both $x = \sqrt[3]{7 - 3x}$ and $x = \dfrac{7 - x^3}{3}$. **[2]**

(c) Use $x_{n+1} = \sqrt[3]{7 - 3x_n}$ with $x_1 = 1.5$ to find α correct to 3 decimal places, showing the result of each iteration. **[3]**

(d) By considering the gradient of $\mathrm{g}(x) = \dfrac{7 - x^3}{3}$ near α, explain why the iteration $x_{n+1} = \dfrac{7 - x_n^3}{3}$ does not converge to α from a starting value close to it. **[3]**

2 Consider the curve $y = xe^{2-x}$ for $x \geqslant 0$, which has a single turning point.

(a) Show that the x-coordinate of the maximum point is exactly 1. **[3]**

(b) Use the sign change method to show that the equation $xe^{2-x} = 1.5$ has a root between 2.5 and 2.6. **[2]**

(c) In this question you must show detailed reasoning. Find the exact area of the region bounded by the curve, the x-axis and the line $x = 1$. **[5]**

Question 1

(a) $\mathrm{f}(x) = x^3 + 3x - 7$: $\mathrm{f}(1) = -3 < 0$ and $\mathrm{f}(2) = 7 > 0$, and f is continuous, so a root lies in $(1, 2)$

(b) $x^3 = 7 - 3x$ gives $x = \sqrt[3]{7 - 3x}$, and $3x = 7 - x^3$ gives $x = \dfrac{7 - x^3}{3}$

(c) $x_2 = 1.3572$, $x_3 = 1.4307$, $x_4 = 1.3938$, $x_5 = 1.4126$, $x_6 = 1.4031$, $x_7 = 1.4079$, $x_8 = 1.4055$, and the values converge to $\alpha = 1.406$

(d) $\mathrm{g}'(x) = -x^2$, and near $\alpha \approx 1.41$, $|\mathrm{g}'(\alpha)| \approx 1.98 > 1$, so the iterates move away from α at each step and do not converge. (For the cube-root form, $|\mathrm{g}'| = \dfrac{1}{x^2} \approx 0.51 < 1$, which is why that one converges.)

Question 2

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{2-x} - x\mathrm{e}^{2-x} = (1-x)\mathrm{e}^{2-x}$, which is zero only at $x = 1$; it is positive for $x < 1$ and negative for $x > 1$, so $x = 1$ is the maximum

(b) $\mathrm{f}(x) = x\mathrm{e}^{2-x} - 1.5$: $\mathrm{f}(2.5) = 0.016 > 0$, $\mathrm{f}(2.6) = -0.073 < 0$, so there is a sign change and a root in $(2.5, 2.6)$

(c) $\displaystyle\int_0^1 x\mathrm{e}^{2-x}\mathrm{d}x = \mathrm{e}^2\left[-x\mathrm{e}^{-x}\right]_0^1 + \mathrm{e}^2\int_0^1 \mathrm{e}^{-x}\mathrm{d}x = \mathrm{e}^2\left(-\mathrm{e}^{-1}\right) + \mathrm{e}^2\left(1 - \mathrm{e}^{-1}\right) = \mathrm{e}^2 - 2\mathrm{e}$